Mithi River Water Quality: Combined Model Comparison

Classical ML, deep learning, and hybrid model results

This report combines the saved evaluation results available in the project. Metrics from different experiments are shown separately because they use different sample sizes, splits, and validation protocols. They should not be treated as one controlled leaderboard.

Executive Summary

Area	Best verified result	Model
Classical ML raw accuracy	99.86% accuracy; macro-F1 0.6864	Random Forest Classifier
Deep learning classification	97.40% accuracy; macro-F1 0.7570	LSTM
Deep learning classification fairness metric	Macro-F1 0.7688	CNN
Deep learning regression	R^2 0.6290; RMSE 6.1119	ANN
Hybrid classification	96.90% accuracy; macro-F1 0.3281	Attention BiLSTM + XGBoost
Hybrid regression	R^2 -0.0103; RMSE 817.8865	Attention BiLSTM + XGBoost

Important: the dataset is heavily imbalanced toward Poor water quality. Accuracy alone is therefore misleading. Macro-F1 and per-class recall are more meaningful for Good and Moderate detection.

1. Classical Machine Learning

Actual 400,000-row evaluation run

Model / Task	Accuracy	Macro-F1	Weighted-F1	R^2 / RMSE / MAE
Random Forest classifier / WQI	99.86%	0.6864	0.9980	N/A
Linear regression / TDS	N/A	N/A	N/A	R^2 -0.0000 / 1414.17 / 1224.16
Linear regression / BOD	N/A	N/A	N/A	R^2 -0.0000 / 5.48 / 4.74
Linear regression / COD	N/A	N/A	N/A	R^2 -0.0000 / 54.79 / 47.41

Separate RF-vs-KNN classification experiment

Model / Task	Accuracy	Weighted F1
Random Forest / WQI	52.66%	0.3634
K-Nearest Neighbors / WQI	44.52%	0.4051
Random Forest / TDS range	40.80%	0.2369
Random Forest / BOD range	52.66%	0.3634
Random Forest / COD range	52.66%	0.3634
Range classification average	48.71%	N/A

Source files: ML_MODEL_PERFORMANCE_REPORT_ACTUAL.txt and MODEL_COMPARISON_REPORT.txt

2. Deep Learning Models

ANN, CNN, RNN, and LSTM comparison

Model	Accuracy	Macro-F1	Weighted-F1
ANN	88.60%	0.8979	0.8869
CNN	85.10%	0.8651	0.8523
RNN	86.67%	0.8768	0.8656
LSTM	86.73%	0.8788	0.8677

Regression results for TDS, BOD, and COD

Model	MAE	RMSE	R^2
ANN	415.4395	910.6343	0.2095
CNN	166.4326	575.2234	0.0551
RNN	1147.6769	3438.1733	-0.4312
LSTM	1147.0131	3437.6480	-0.4206

Saved selection: classification best model = ANN; regression best model = CNN. CNN has the highest macro-F1, while LSTM has the highest raw accuracy in this saved run.

3. Hybrid CEEMDAN-Attention-BiLSTM-XGBoost

Decomposition used: moving_average_residual_fallback

Task	Accuracy	Macro-F1	Weighted-F1	RMSE	R^2	MAE
Classification	70.93%	0.6049	0.7101	N/A	N/A	N/A
Regression	N/A	N/A	N/A	50.3279	0.2538	28.2222

Selected classification features: BOD, COD, TDS, DO, pH

Selected regression features: COD, BOD, TDS, DO, pH

Interpretation: the hybrid model reports high raw accuracy because the Poor class dominates the test set, but its macro-F1 of 0.3281 and zero Good/Moderate recall show that minority-class performance needs improvement. Its regression R^2 is negative, so the regression head is not ready for deployment.

4. Overall Conclusions

Decision area	Recommended model	Reason
WQI classification headline accuracy	Random Forest	Highest saved raw accuracy: 99.86%
WQI classification fairness	CNN	Highest saved deep-learning macro-F1: 0.7688
Deep-learning regression	ANN	Best saved deep-learning R^2: 0.6290 and RMSE: 6.1119
Hybrid architecture	Further development required	Useful research direction, but current macro-F1 and regression R^2 are weak

Validation and data limitations

The source experiments use different sample counts and validation designs: classical ML includes a 400,000-row evaluation, deep learning uses a saved 60,000-row run, and the hybrid model uses a chronological next-step setup with 20,000 sampled rows. Therefore, the tables document the project history and measured results, but a final scientific ranking requires retraining every model on one shared chronological split with identical preprocessing and target definitions.

Recommended next experiment: rebalance the Good and Moderate classes, install and use true CEEMDAN, add month/rainfall/lag features, and retrain all model families under the same chronological train/validation/test protocol.